AgNova

AI-Powered Autonomous Crop Care Solution

Integrating robotics, agronomic intelligence and biological crop inputs

A next-generation agricultural platform that continuously scans fields, diagnoses crop stress, and delivers precision biological treatments.

Presented by
Kriya AgTech
(A division of Kriya Biosys)

Agriculture Is Entering a New Era

But problems remain the same

- Increasing pest and disease pressure
- Pesticide resistance spreading globally
- Soil degradation and declining fertility
- Stricter regulations on chemical crop protection
- Demand for sustainable farming practices

Agriculture requires smarter crop care systems.

The Core Problem

Detection
Crop issues are often discovered too late.

Monitoring
Large farms make manual monitoring impractical.

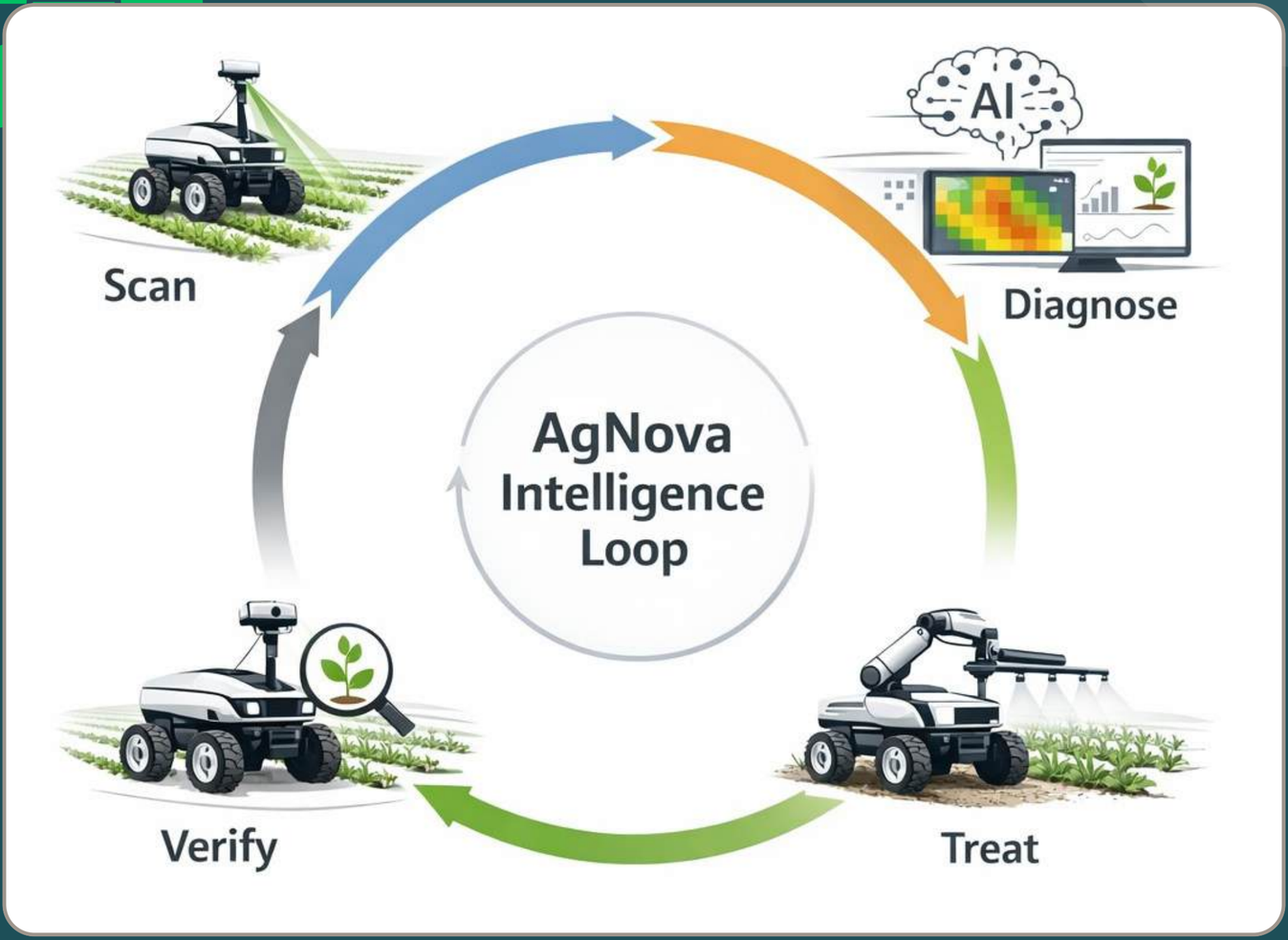
Treatment
Most interventions rely on blanket spraying.

The Precision Agriculture Gap

But treatment decisions remain manual.

Introducing AgNova

An integrated platform combining robotics, AI and biological crop inputs.

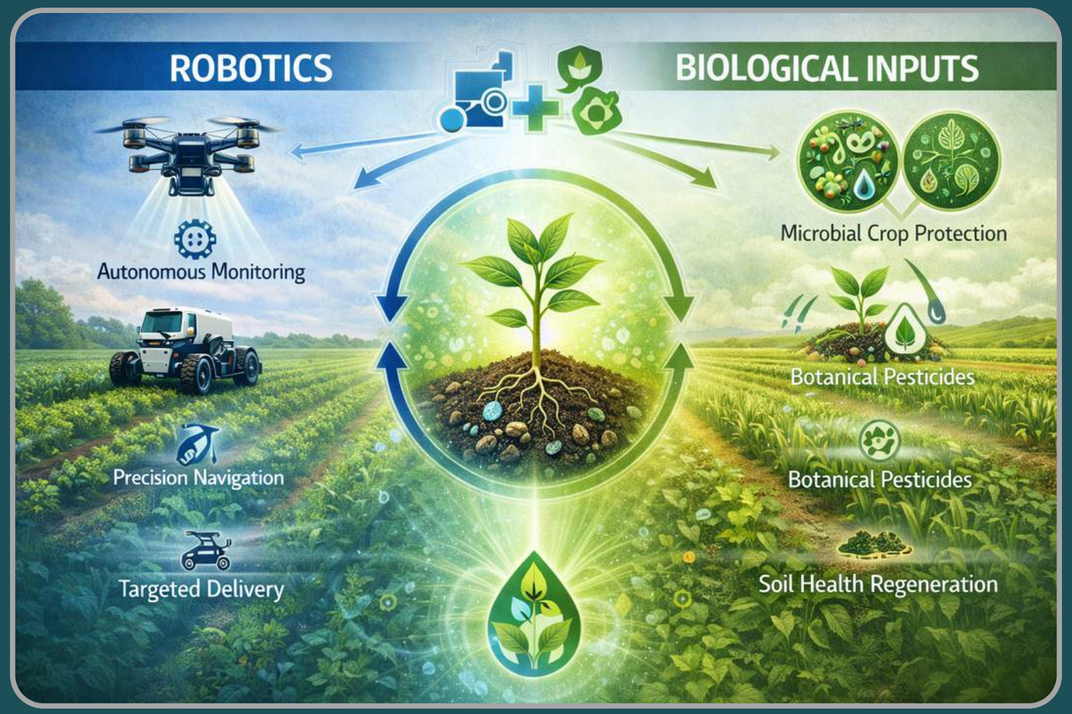


Scan → Diagnose → Treat → Verify

AgNova = Robotics + Biologicals

Robotics

- Autonomous monitoring
- Precision navigation
- Targeted delivery



Biological Inputs

- Microbial crop protection
- Botanical pesticides
- Soil health regeneration

AgNova operates at the convergence of these technologies.

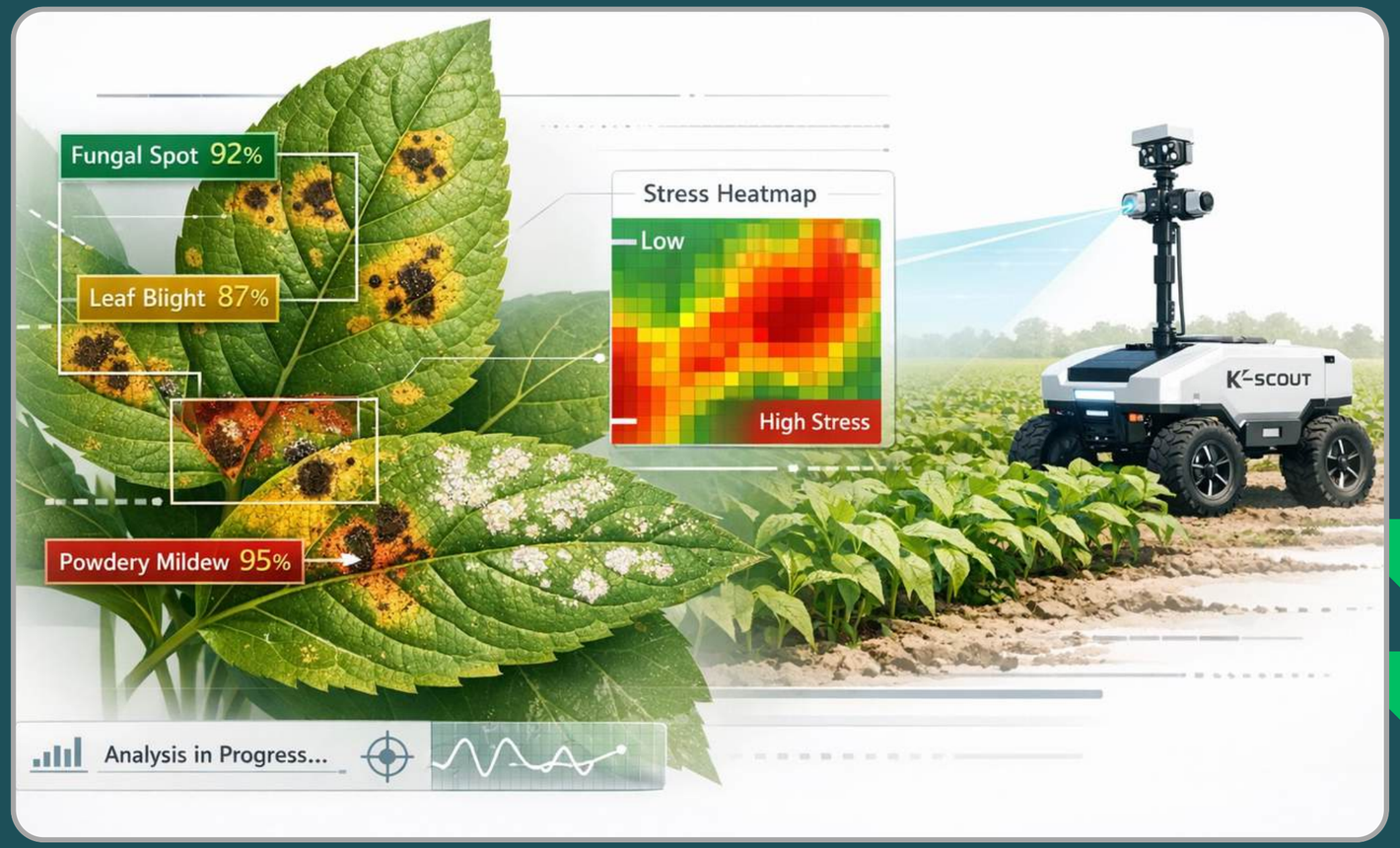
Platform Architecture

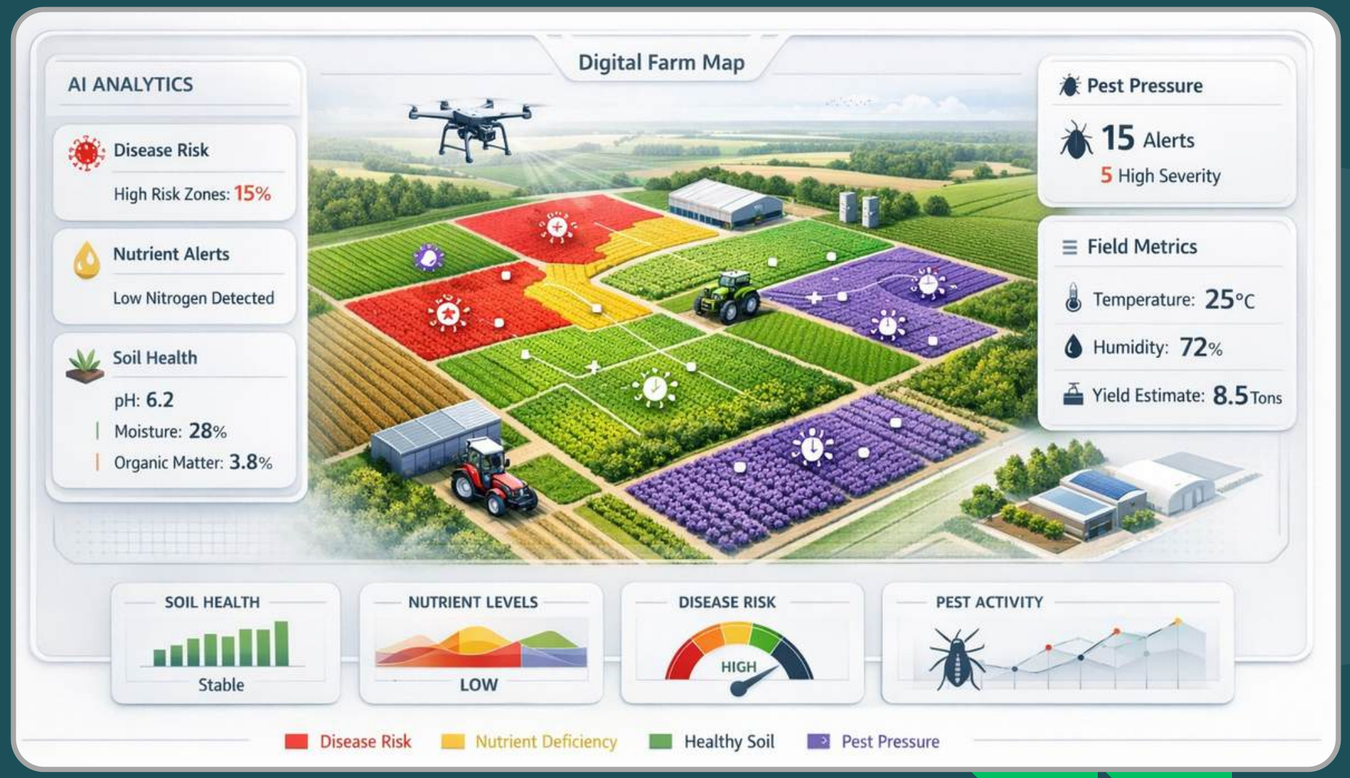
Operational Workflow

- Continuous field scanning
- AI agronomic analysis
- Precision biological intervention
- Verification and learning

K-Scout Rover

- RTK GPS navigation
- RGB + multispectral imaging
- Thermal stress detection
- Soil sensing and microclimate monitoring





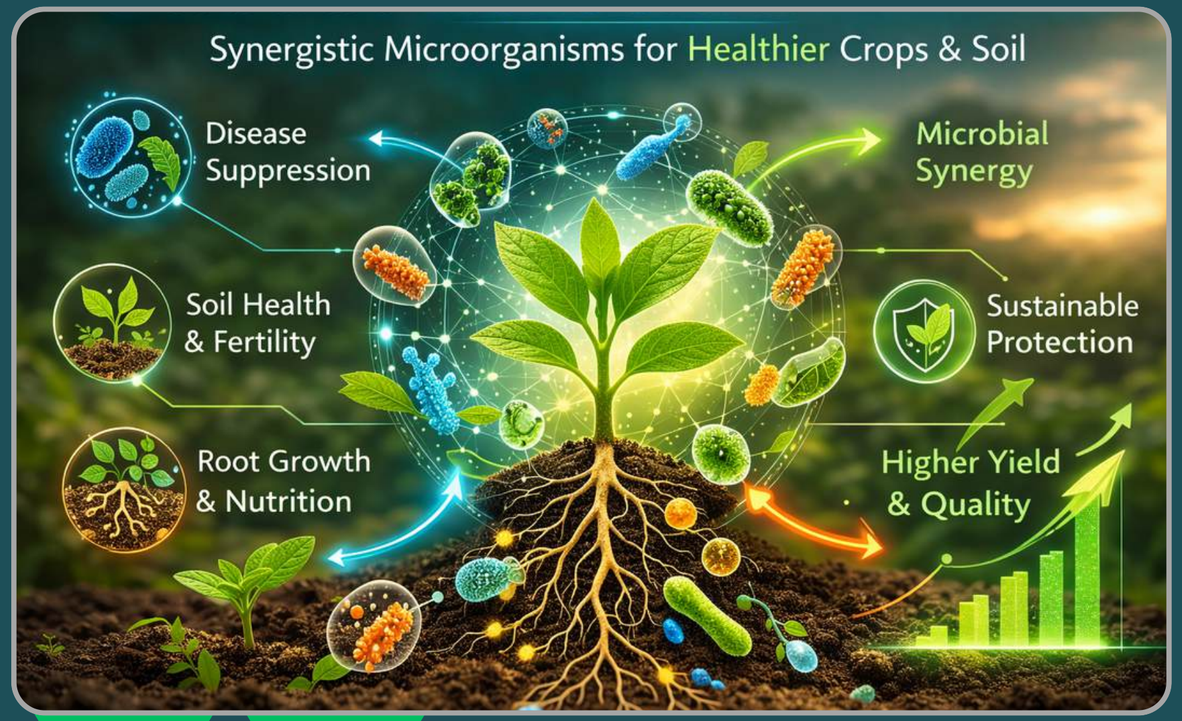
Kriya Ag Intelligence

- Disease detection models
- Pest pressure analysis
- Nutrient deficiency diagnostics
- Predictive agronomy analytics

K-Fend Rover

- Precision micro-spraying
- Root-zone microbial injection
- Localized soil drench
- Plant-level treatment targeting

Kriya™



Dynamic Biological Consortia

1

Microbial Inputs

- Trichoderma
- Bacillus
- Beauveria

2

Botanical Inputs

- Neem
- Essential oils
- Plant extracts

3

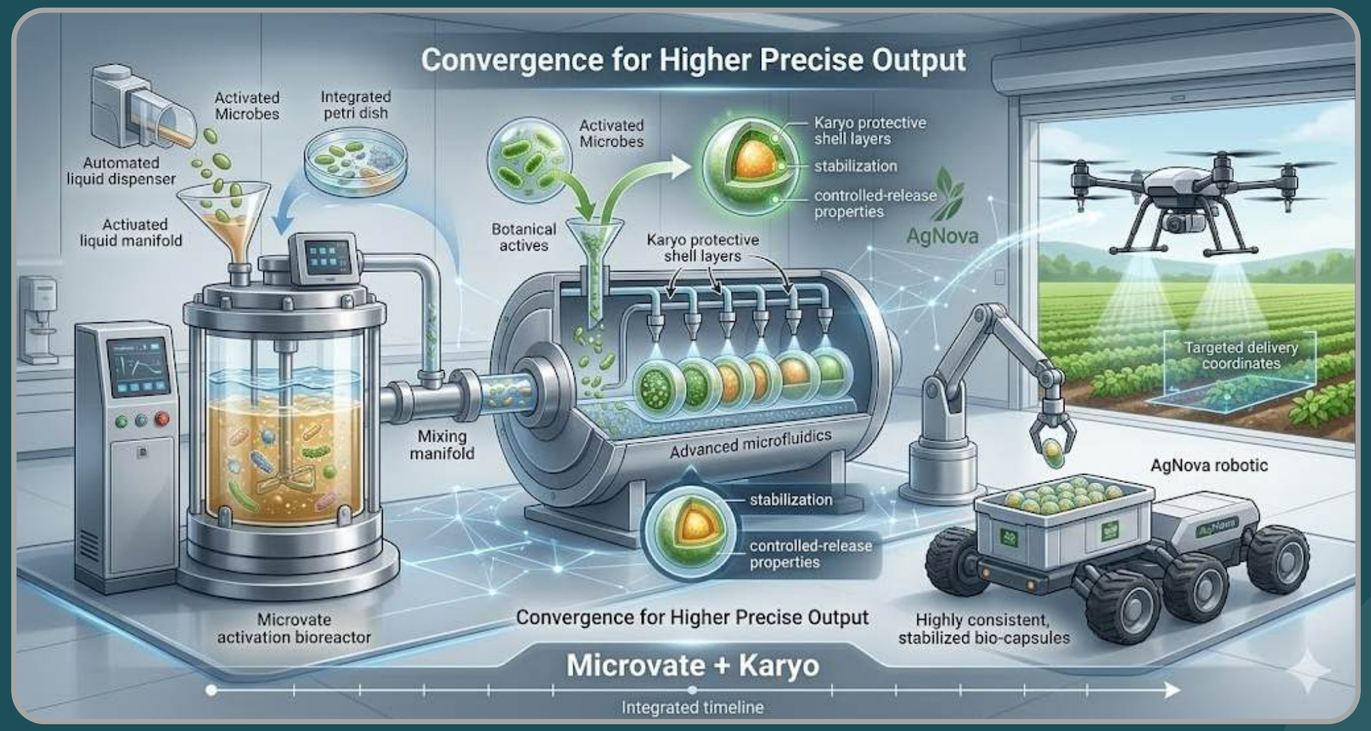
Biostimulants

- Seaweed
- Humic substances

Microvate + Karyo

Microvate

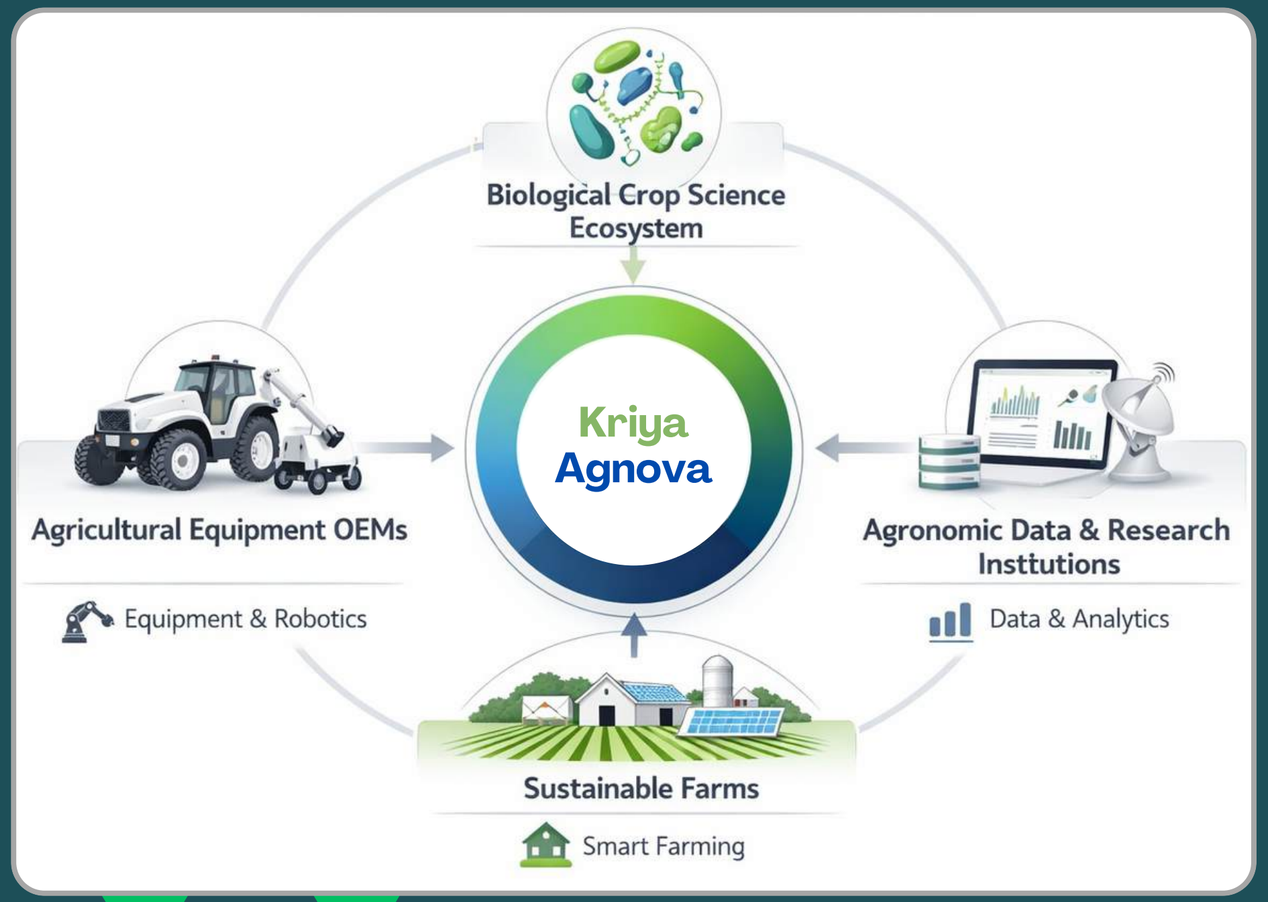
Just-in-time microbial activation



Karyo

Encapsulation and stabilization technology

Maximizing biological effectiveness in the field.



Technology Moat

1. Robotics Platform
2. Agronomic AI
3. Biological Technologies
4. Agricultural Data

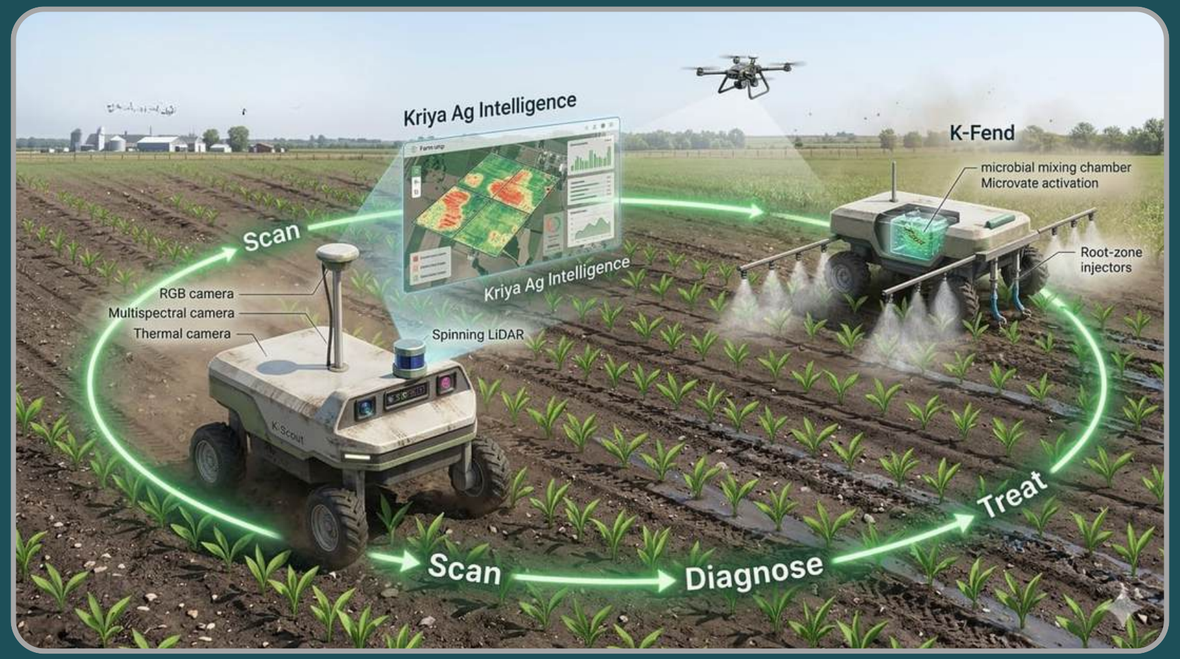
Strategic Partnerships

- Agricultural equipment OEMs
- Biological crop science ecosystem
- Agronomic research institutions
- Farm deployment partners

Product Development Roadmap

- Autonomous field intelligence platform
- Agronomic AI expansion
- Precision biological treatment systems
- Integrated robotic crop care
- Autonomous biological agriculture

The Future of Crop Care

AgNova

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